Warsaw University of Technology
Poland

Controlling Uncertainty and Hallucination Risk in Multi-Agent Fact Verification

Adam Kostka · Jarosław A. Chudziak
adam.kostka.stud@pw.edu.pl

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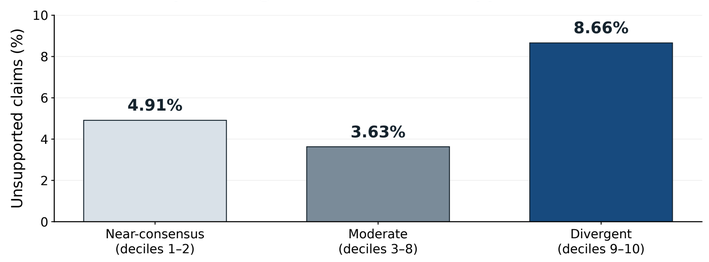
How should uncertainty be measured in multi-agent fact verification?

Multi-agent systems can combine responses from several agents to the same grounded query. The relationship between agreement among those responses and factual reliability may be more complex than a direct correspondence. Alignment among complete responses and variation in claim support scores are therefore examined as complementary signals for selecting claims under a risk constraint.

1. Why Multi-Agent Agreement Is Not Enough

Controlling uncertainty and hallucination risk requires determining whether multi-agent agreement is a reliable signal of factual correctness. Complete responses from five agents based on the same model were compared by alignment, while their extracted claims were assessed for support in the grounding document.

How response alignment relates to unsupported claims



The observed non-monotonic association suggests that response alignment can inform uncertainty assessment, but not through a simple relation in which greater agreement implies greater reliability. Elevated rates of unsupported claims are observed both when responses nearly agree and when they differ strongly.

2. Research Questions and Hypothesis

In applications where unsupported claims matter, correlated failures among agents can complicate their detection. At the same time, overly conservative filtering can remove claims that are supported by the grounding document. Together with the observed non-monotonic relationship between response alignment and factual error, this trade-off motivates two research questions about assessing risk while retaining supported claims.

Research Question 1

Which uncertainty-aware scoring mechanism retains the most supported claims under the same risk limit?

Research Question 2

How can hallucination risk be controlled when errors within a multi-agent query are correlated?

Hypothesis

The relationship between response alignment and factual error is non-monotonic. A scoring mechanism that also accounts for variation in claim-support scores should therefore balance risk control and claim retention more effectively than agreement or raw support scores alone.

These questions motivate a claim-level pipeline that combines support scoring, uncertainty adjustment, and calibrated selection.

3. Uncertainty-Aware Verification Pipeline

Each agent receives the same grounding document and query. The resulting responses are decomposed separately into verifiable claims, pooled within a single query-response tuple, and assigned factual-support scores by a weak support judge.

A. Grounded input

Grounding document (excerpt)
Talking 2-XL Robot by Tiger Electronics User Reviews
1. “Great toy. Still a hit with the next generation of kids as well.”
2. “It was perfect, exactly what I had hoped it would be!”
3. “Just what I expected.”

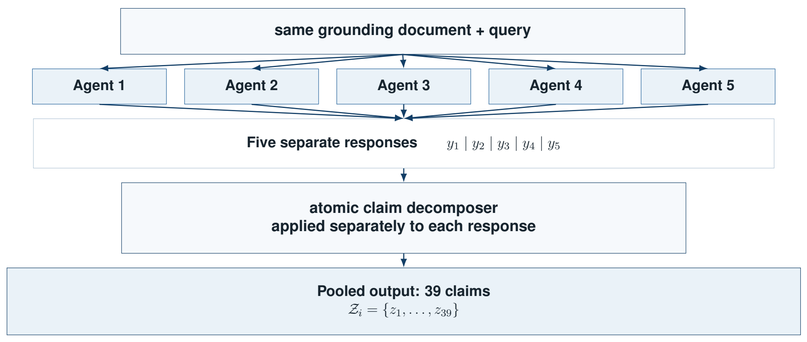
Query: What are the primary positive aspects reviewers refer to?

One query-response tuple

A query-response tuple groups one grounded query/reference pair with all claims extracted from its five agent responses. Here, five responses produce 39 pooled claims that are treated together as one statistical unit.

5 responses ⟶ 39 claims ⟶ 1 tuple

B. Generation and claim extraction



C. Weak support scoring



Example claims and weak scores

Claim	Weak score
General satisfaction.	.95
Described as “perfect.”	.75
Met reviewers’ expectations.	.88

4. Measuring Ensemble Uncertainty

Ensemble uncertainty is measured at two complementary levels. Complete agent responses are represented by vector embeddings, whose pairwise cosine similarities form an ensemble similarity matrix. Spectral entropy H_{spec} summarizes the alignment structure of this matrix: lower values indicate stronger agreement, while higher values indicate greater response diversity. For claim selection, a distinct quantity, σ_{tuple}, measures variation among the weak factual-support scores assigned to all claims in the same query-response tuple.

Claim-support variation

$$\sigma_{\text{tuple}} = \sqrt{\frac{1}{M} \sum_{m=1}^{M} (u_m - \bar{u})^2}$$

Deviation-adjusted claim score

$$S(z_m) = \frac{u_m}{1 + \gamma \sigma_{\text{tuple}}}$$

For claim z_m, u_m denotes its weak factual-support score, M is the number of claims in the tuple, $\bar{u}$ is their mean weak score, γ controls the strength of the tuple-level adjustment, and $S(z_m)$ is the resulting deviation-adjusted score.

Worked example

For $u_m = .700$, $\sigma_{\text{tuple}} = .2766$, and $\gamma = .5$:

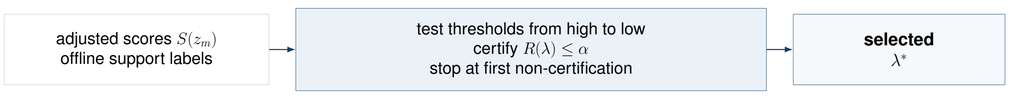
$$S(z_m) = \frac{.700}{1 + .5(.2766)} = .6149$$

Greater support-score variation lowers the adjusted claim score. Section 5 calibrates this score, and Evaluation 1 compares it with raw support and entropy-based alternatives.

5. Controlling Risk: Learn-Then-Test

False discovery proportion (FDP) is the fraction of retained claims that are unsupported within one query-response tuple. Its expectation across exchangeable tuples defines the false discovery rate, $R(\lambda) = \mathbb{E}[\text{FDP}(\lambda)]$. Learn-Then-Test uses adjusted scores and offline support labels to select the least conservative threshold λ^* that satisfies the target risk level.

Offline calibration



Fixed-sequence Learn–Then–Test with combined Hoeffding–Bentkus p-values.

$$\Pr(R(\lambda^*) \le \alpha) \ge 1 - \delta$$

Here, α is the target expected FDR and δ is the allowed calibration failure probability.

Deployment



Applying the calibrated threshold. Section 4 gives $S(z_m) = .6149$. At $\alpha = .05$, calibration selects $\lambda^* = .6231$; since $S(z_m) < \lambda^*$, the claim is not retained. Offline support labels are used only for calibration, while deployment applies the fixed threshold to new scores. The guarantee controls expected FDR across exchangeable query-response tuples, not each individual query.

6. Experimental Design and Evaluation Protocol

Three evaluations address the research questions through penalty comparison, validation of risk control, and robustness analysis.

Benchmark: FACTS Grounding | Sample: 300 tuples / 22,155 claims | Ensemble: 5 homogeneous Gemini 2.5 Flash agents | Split: 200 calibration / 100 test | Supervision: weak-judge scores / offline strong-judge labels

Evaluation 1: Penalty Comparison

Test. Compare three scoring configurations using the same LTT procedure and risk targets: raw support (unadjusted weak score), monotonic entropy (low-entropy penalty), and Score Deviation (score variation).

Measure. Mean supported-claim recall over 20 reshuffled calibration–test splits at the same target FDR.

Evaluation 2: Risk Control

Test. Apply the Score Deviation threshold selected during calibration to held-out query-response tuples and compare empirical FDR with its target.

Measure. Empirical tuple-level FDR with 95% bootstrap intervals relative to the target-risk boundary.

Evaluation 3: Mechanism and Robustness

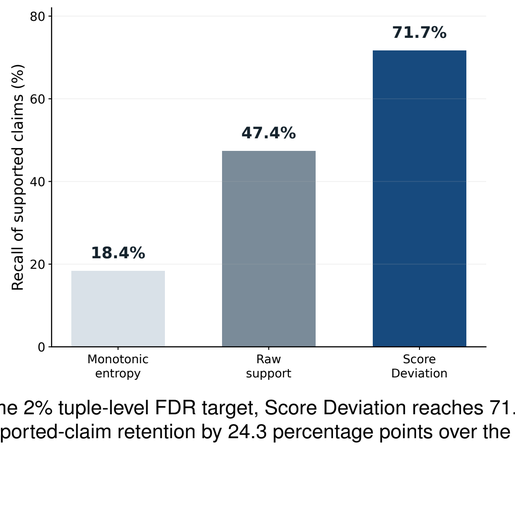
Test. Stratify tuples by entropy, vary the penalty weight, repeat calibration across reshuffled splits, and evaluate a matched mixed-model ensemble.

Measure. Error pattern, recall, FDR, and threshold stability.

7. Experimental Results

Evaluation 1

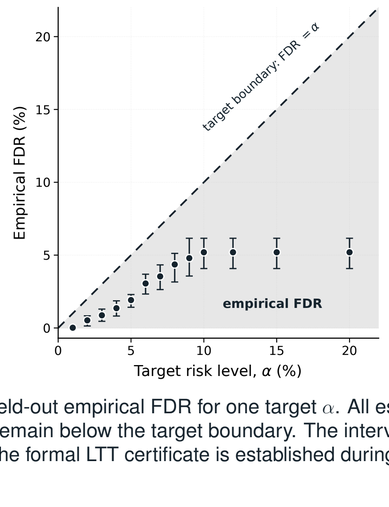
Score Deviation retains the most supported claims



Under the same 2% tuple-level FDR target, Score Deviation reaches 71.7% recall, improving supported-claim retention by 24.3 percentage points over the raw support score.

Evaluation 2

The additional retained output remains risk-controlled



Each point shows held-out empirical FDR for one target α. All estimates and 95% bootstrap intervals remain below the target boundary. The intervals provide an empirical check, while the formal LTT certificate is established during calibration.

Evaluation 3

The mechanism remains stable across penalty weights, repeated splits, and generator ensembles

Penalty weight	At $\alpha = .05$, recall saturates beyond $\gamma \approx .5$ while FDR remains between .029 and .034.
Repeated splits	All 20 calibration splits satisfy the $\alpha = .05$ target.
Degenerate folds	At $\alpha = .02$, 0/20 with Score Deviation, compared with 14/20 for raw support.
Heterogeneous generators	Matched 100-query test at $\alpha = .05$: recall .920 vs. .924; FDR .0148 vs. .0146 (heterogeneous vs. homogeneous).
Enriched base rate	10.85% unsupported: recall .630 and FDR .0121 at $\alpha = .05$.

Mechanism. The non-monotonic alignment-error pattern in Section 1 explains why a monotonic entropy penalty underperforms. Score Deviation remains effective across penalty weights, calibration splits, and initial generator composition and base-rate stress tests.

8. Discussion, Limitations, and Future Work

Score Deviation is most useful at strict risk budgets, where unadjusted scores can yield total abstention. Initial matched heterogeneous generator and enriched FACTS stress tests also preserved FDR control. However, model diversity did not remove unsupported claims, and the enriched setting became conservative at stricter budgets. These are initial robustness results, not broad validation.

Limitations

- Evidence remains limited to FACTS; the mixed-generator and enriched-base-rate results are stress tests, not cross-domain validation.
- The certificate is relative to strong-judge labels; the full experiment used about 22,000 such labels. Label bias and calibration cost remain deployment constraints.
- Exchangeability is required. Model, judge, prompt, domain, or query distribution shifts require new labels and recalibration.

Future Work

- Test larger heterogeneous ensembles, more datasets and domains, and alternative weak and strong judges.
- Reduce calibration label requirements and validate judge labels against human annotations.
- Develop weighted or online recalibration for non-stationary deployment streams.

9. Key Takeaways

The results support three practical conclusions about the limits of agreement, the appropriate level of uncertainty measurement, and the role of calibration in risk control.

- **Agreement does not guarantee factual reliability.**
  In the evaluated homogeneous ensemble, unsupported claims occur more frequently at both near-consensus and strong-divergence extremes. Multi-agent agreement is therefore an incomplete measure of uncertainty.
- **Uncertainty must be measured at the appropriate level.**
  Spectral entropy H_{spec} summarizes alignment among complete agent responses, while σ_{tuple} measures variation among claim-support scores. The latter provides the operational signal used by Score Deviation.
- **Calibration converts uncertainty into risk control.**
  Learn–Then–Test converts adjusted scores into a constraint on expected FDR. At $\alpha = .02$, Score Deviation increases supported-claim recall from 47.4% to 71.7% under the same risk target. Initial heterogeneous generator and enriched-base-rate stress tests also remained risk-controlled, although broader validation remains open.